

# NATIONAL TELE-OPHTHALMOLOGY SCREENING NETWORK

Automated Diabetic Retinopathy Diagnostic Validation Report — Rural PHC Deployment

## 1. PATIENT DEMOGRAPHICS & CLINICAL PROFILE

|                   |                |                        |                       |
|-------------------|----------------|------------------------|-----------------------|
| Patient Name:     | Arslan sa      | Screening Session ID:  | fc12e302-a04c-4c8a... |
| Age / Gender:     | 45 yrs / Male  | Contact Phone:         |                       |
| Diabetes History: | Type 2 (5 yrs) | Screening Date & Time: | 2026-09-22 18:01:11   |

## 2. AI MULTI-MODAL DIAGNOSTIC FINDINGS (ICDR CRITERIA)

ICDR Diagnosis: Moderate NPDR (Grade 2 - Exudates & Hemorrhages) | Calibrated Confidence: 94.2% | Image Quality: BORDERLINE / MEDIUM

Recommended Clinical Directive: Ophthalmologist Referral within 4-6 Weeks

## 3. QUANTITATIVE LESION & VASCULAR BIOMARKERS

| Biomarker Structure              | Count / Value | Clinical Pathology (ICDR Classification)            |
|----------------------------------|---------------|-----------------------------------------------------|
| ■ Microaneurysms / Hemorrhages   | 0             | Focal vascular outpouchings (Stage 1+ indicator)    |
| ■ Hard Exudates (Lipid Residue)  | 7             | Lipoprotein leakage marker (Moderate NPDR Stage 2+) |
| ■ Cotton Wool Spots (Ischemia)   | 0             | Axoplasmic flow blockage (Severe NPDR Stage 3+)     |
| ■ Retinal Vessel Network Density | 65.31%        | Normal reference baseline: 8.0% - 18.0%             |
| ■ Optic Disc Landmark Coordinate | (136, 228)    | Verified reference center for NVD exclusion         |

## 4. AUTHORIZING OPHTHALMOLOGIST EVALUATION & SIGN-OFF

|                                                                                                                      |                                                                                             |
|----------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| Authoring Doctor: Dr Manushi                                                                                         | Specialization: Senior Vitreo-Retina Consultant                                             |
| Medical License Reg No: MCI-RET-1680                                                                                 | Hospital Affiliation: Krishna                                                               |
| Clinical Status: Pending Review                                                                                      | Validation Timestamp: None                                                                  |
| Doctor Prescriptions & Directives:<br>Verified presence of focal macular hard exudates. Routine follow-up scheduled. | Official Stamp & Digital Signature:<br><i>Signed Electronically by Authorized Clinician</i> |

\* Generated by Netra Setu Tele-Ophthalmology Network. Smart India Hackathon (SIH 2026). Stored permanently in MongoDB Atlas Cloud Cluster.